Nimbus Owner's Manual

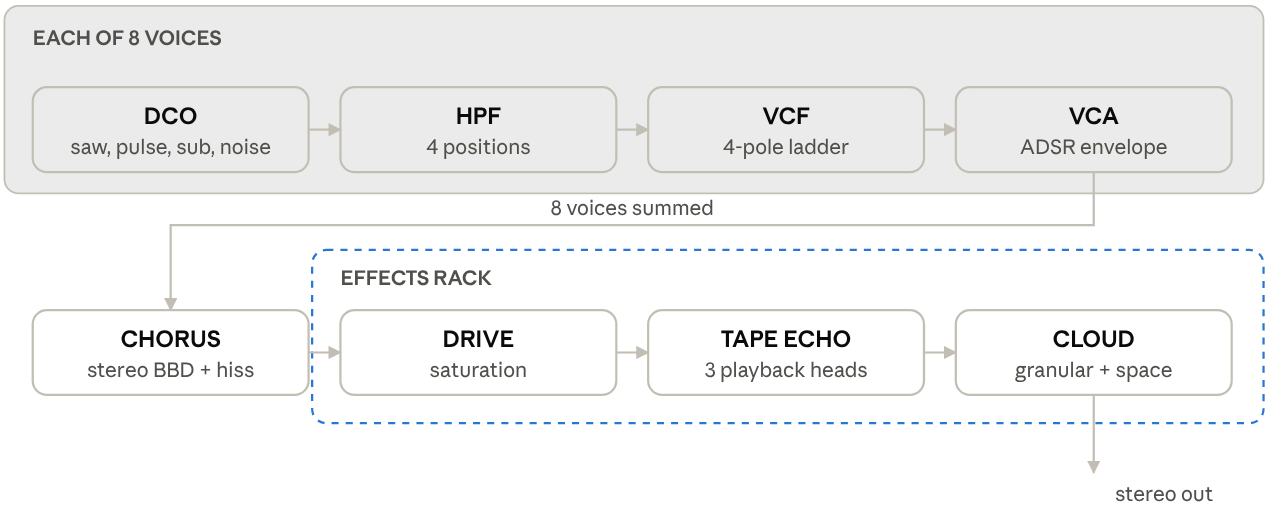
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Welcome

Nimbus is an 8-voice pad synthesizer modelled on the single-oscillator (DCO) polysynths of the early 1980s, with a built-in effects rack of drive, tape echo and a granular CLOUD section. It runs as a VST3 or Audio Unit plug-in in Ableton Live and other DAWs, and as a standalone app.

Every voice has one DCO: a saw, a pulse with pulse-width modulation, a sub square one octave down, and white noise. The voice then passes through a four-position high-pass filter, a 4-pole ladder low-pass filter and an envelope-controlled amplifier. All voices meet in a stereo bucket-brigade chorus, which brings its own hiss. Then the mix runs through the effects rack, in a fixed order: DRIVE, then TAPE ECHO, then CLOUD.

Each voice is shaped on its own, then all eight share the chorus and effects



Nimbus signal flow · per-voice path, shared chorus and effects rack

The same DCO engine also runs on the KORG microKORG2 as a hardware oscillator (see the last section).

Installing

Nimbus needs macOS 11 or later and runs natively on Apple Silicon and Intel Macs. The installer comes as a disk image, `Nimbus-<version>-macOS.dmg`, from the project's Releases page on GitHub.

1. Open the `.dmg` and double-click **Install Nimbus.pkg**.
2. macOS says it can't verify the developer, because this build isn't notarized by Apple. Click **Done**, open **System Settings → Privacy & Security**, scroll down and click **Open Anyway** next to the installer, then enter your password.
3. Follow the installer. Click **Customize** to choose which parts to install:

Component	Installed to	Used by
VST3 plug-in	/Library/Audio/Plug-Ins/VST3	Ableton Live and most DAWs
Audio Unit	/Library/Audio/Plug-Ins/Components	Ableton Live, Logic Pro, GarageBand
Standalone app	/Applications	playing without a DAW

The installer also removes any copies of Nimbus you put in your own `~/Library/Audio/Plug-Ins` folders by hand, so your DAW never sees two Nimbus plug-ins. Your saved presets are never touched.

Loading Nimbus in Ableton Live

1. Open **Settings → Plug-Ins**.
2. Turn on **Use VST3 Plug-In System Folders**, **Use Audio Units v2**, or both.
3. Click **Rescan**.
4. In the browser, open **Plug-Ins → Javi** and drag **Nimbus** onto a MIDI track.

If both formats are enabled, Nimbus appears twice, once as VST3 and once as AU. They sound identical; pick one and use it consistently, because a Live set remembers which format it used.

The synth panel

The top two rows are the synthesizer; each coloured header names a section. Drag a fader to change it, or click the number under it to type a value. Every control can be automated and MIDI-mapped from your DAW. In Ableton, click **Configure** on the device title bar, then move a control to add it.

DCO (orange)

Control	Range	What it does
PW	0–100	Pulse width, from a square (0) to a thin 5% pulse (100).
PWM	0–100	How far the built-in triangle LFO sweeps the pulse width. This is the classic slowly breathing pad sound.
RATE	0–100	PWM LFO speed, about 0.1 Hz to 19 Hz.
SAW	0–100	Saw level.
PULSE	0–100	Pulse level.
SUB	0–100	Square wave one octave down, locked in phase to the DCO. Adds weight without detuning.
NOISE	0–100	White noise into the mixer, for breath and air.

HPF (yellow)

Four positions, as on the original hardware: **0** boosts the bass below about 100 Hz, **1** is flat, **2** cuts below about 240 Hz and **3** below about 720 Hz. Use 2 or 3 to keep a pad out of the way of bass and kick.

VCF (red)

Control	Range	What it does
FREQ	20 Hz–18 kHz	Cutoff of the 4-pole (24 dB/octave) ladder low-pass filter.

Control	Range	What it does
RES	0–100	Resonance. Near the top the filter starts to ring and whistle. Some low end is restored automatically so the sound doesn't go thin.
ENV	0–100	How far the envelope opens the filter, up to 7 octaves.
KYBD	0–100	Key tracking: at 100 the cutoff follows the keyboard exactly, at 0 every note gets the same cutoff.

ENV (teal)

One ADSR envelope drives both the filter (through ENV) and the volume of each voice. **A** attack runs from 1 ms to 10 s, **D** decay from 5 ms to 10 s, **S** sustain from 0 to 100, and **R** release from 5 ms to 12 s.

LFO (green)

One triangle LFO for the whole synth, as on the original.

Control	Range	What it does
RATE	0.1–20 Hz	LFO speed.
DELAY	0–5 s	How long after each note the LFO takes to fade in. Great for vibrato that arrives late, like a string player's.
PITCH	0–100	Vibrato depth, up to one semitone either way.
VCF	0–100	Filter sweep depth, up to two octaves either way.

CHORUS (blue)

OFF, **I**, **II** or **I+II**. Two bucket-brigade delay lines are swept in opposite directions by one LFO, one feeding the left channel and one the right. **I** is slow and gentle (0.51 Hz), **II** is deeper (0.86 Hz), and **I+II** is a fast, shallow shimmer (9.75 Hz). This chorus is most of what makes the sound wide and lush.

CHARACTER (grey)

Control	Range	What it does
HISS	0–100	Bucket-brigade hiss, always present like on the real circuit, and louder when the chorus is on. Set it to 0 for a clean sound.
DRIFT	0–100	Each note lands slightly off pitch and wanders slowly, up to about ±9 cents.
AGE	0–100	An ageing-circuit macro: per-voice saturation, a faint 60 Hz mains hum and a duller top end.
SPREAD	0–100	Pans voices alternately left and right for a wider spread of notes. At 0 every voice sits in the centre.

OUT

VOL sets the synth's level from −36 dB to +6 dB, before the effects rack.

The effects rack

DRIVE sits at the right of the second row and the other two fill the bottom row. They run in a fixed chain: DRIVE → TAPE ECHO → CLOUD. Each has an **ON/OFF** switch in its header; switched off, it is bypassed completely. Switching TAPE ECHO or CLOUD back on starts it with an empty buffer, so old echoes never jump back in.

DRIVE

Gentle tube-style saturation that adds warmth and a little grit while keeping the level roughly constant.

Control	Range	What it does
DRIVE	0–100	Amount of saturation. Up to about 40 it stays subtle; above that it starts to growl.
TONE	0–100	Brightness of the driven sound, from 1.5 kHz to 18 kHz.
MIX	0–100	Blend of clean and driven sound.

TAPE ECHO

A three-head tape delay in the spirit of the classic space echo units. The sound is recorded to a loop of virtual tape and played back by three heads at 1×, 2× and 3× the base delay time.

Control	Range	What it does
TIME	0–100	Tape speed. Head 1 runs from 50 ms to 350 ms, so head 3 reaches about 1 second. Moving TIME while echoes play makes them bend in pitch, like a real tape machine changing speed.
INTENS	0–100	Feedback, the number of repeats. Above about 85 the echoes keep building into self-oscillation, held in check by tape saturation.
ECHO	0–100	Level of the echoes.
TONE	0–100	Brightness of the repeats. Low settings sound like old, worn tape.
WOW	0–100	Wow and flutter: slow and fast wobble of the tape speed.
Heads	1, 2, 3, 1+2, 2+3, 1+3, ALL	Which playback heads are active. Head 1 is panned left, head 2 right and head 3 centre.
Sync	Free, 1/16, 1/8T, 1/8, 1/8D, 1/4	Free uses TIME. A note value locks head 1 to your project tempo; heads 2 and 3 follow at 2× and 3×.

CLOUD

A granular micro-looper. CLOUD constantly records the last few seconds of sound and plays it back as many short overlapping fragments called grains. Play a chord and let it ring to hear it bloom.

Mode	What it does
MOSAIC	Grains from random moments in the recent past, some an octave up or down. Turns chords into glittering patterns.
HAZE	Long, smooth, slightly detuned grains spread wide. A soft wash that thickens pads.
GLIDE	Grains that bend up or down by a fourth as they play. Swooping, tape-like movement.
REVERSE	Grains played backwards, some an octave down. Dreamy, ghostly swells.

Control	Range	What it does
ACTIVITY	0–100	How many grains: from about 1.5 to 40 per second.
TIME	0–100	Grain length (40 ms to 600 ms) and how far back in time grains can reach (up to about 3 seconds).
SHAPE	0–100	Grain envelope, from clicky and percussive (low) to smooth and blurred (high).
REPEATS	0–100	Feeds the cloud back into its own recording, so textures build on themselves.
FILTER	0–100	Low-pass filter on the cloud, 400 Hz to 18 kHz.
SPACE	0–100	Built-in reverb on the cloud; higher settings give a longer tail (up to about 5.6 s).
MIX	0–100	Balance of the dry sound and the cloud.
FREEZE	on/off	Stops recording, so the cloud keeps playing what it holds while you play something else on top.

Presets

The bar at the top of the window is the preset browser: **<** and **>** step through presets, the menu between them lists them all, **SAVE** stores the current sound, **DELETE** removes a preset you saved and **FOLDER** opens your preset folder in the Finder.

Saving your own sounds

1. Shape a sound.
2. Click **SAVE**, type a name and press Return. Saving under an existing name replaces that preset.
3. Your preset appears under **USER** in the preset menu, in every project.

User presets are stored as `.nimbuspreset` files in `~/Library/Audio/Presets/Javi/Nimbus`. You can back them up, share them or copy them to another Mac; anything in that folder shows up in the menu. **DELETE** only works on user presets and asks before it removes the file. Factory presets can't be changed, but you can save an edited version under a new name.

Your DAW also remembers Nimbus's exact settings inside each project, whether or not you saved a preset. Ableton's own **Save Preset** button on the device title bar works too.

Factory presets

Preset	Effects on	Sound
Classic Strings	—	The defaults: saw and pulse, PWM, chorus II. The essential string pad.
Tape Dream Pad	Tape echo	Slow-attack pad into all three heads with wobbly, darkened repeats.
Cloud Strings	Cloud (HAZE)	Strings smeared into a wide, reverberant haze.
Mosaic Keys	Cloud (MOSAIC)	Short plucky chords that scatter into octave fragments.
Shimmer Pad	Cloud (GLIDE)	Bright PWM pad, chorus I+II, gliding grains.
Warm Brass Pad	Drive	Saw-only brass with a filter swell and light saturation.

Preset	Effects on	Sound
Hollow PWM	Tape echo (synced)	Thin pulse, deep PWM, dotted-eighth echoes on head 3.
Worn Voice Card	Drive, tape echo	Maximum age: drift, hiss, hum, bass boost, heavy wow.
Night Drive	Tape echo (synced)	Dark resonant pad with a slow filter LFO and quarter-note echoes.
Reverse Memory	Cloud (REVERSE), tape echo	Backwards swells behind the chords.
Vibrato Strings	—	Strings with delayed vibrato that fades in after each note.
Plain DCO	—	No chorus, hiss, drift or age: the bare oscillator, as a starting point.

Sound design recipes

Each recipe starts from **Classic Strings**.

Warm analogue pad

1. Set **A** to around 1 s and **R** to 2–3 s.
2. Lower **FREQ** to about 1.5 kHz and set **ENV** to 20 for a gentle swell.
3. Set **HPF** to **0** for extra low-end weight.
4. Choose chorus **II** and set **DRIFT** to 30.

Breathing PWM choir

1. Set **SAW** to 0 and **PULSE** to 100.
2. Set **PW** to 20, **PWM** to 70 and **RATE** to 15 for a slow sweep.
3. Raise **NOISE** to 5 for breath.
4. Add **LFO** with **DELAY** at 0.8 s and **PITCH** at 15 for vibrato that arrives late.

Dub chord stabs

1. Set **A** to 5 ms, **D** to 300 ms, **S** to 0 and **R** to 200 ms.

2. Turn on **TAPE ECHO**: **Sync** 1/8D, heads **1+3**, **INTENS** 60, **WOW** 40, **TONE** 35.
3. Turn on **DRIVE** at 30 so the repeats get grittier as they build.
4. Ride **INTENS** up past 85 in a breakdown for runaway echoes, then pull it back.

Frozen ambience

1. Turn on **CLOUD** in **HAZE** mode with **SPACE** at 80 and **MIX** at 70.
2. Hold a chord for a few seconds.
3. Press **FREEZE**. The cloud keeps playing that chord while you play a melody over it.
4. Turn **FREEZE** off to start capturing again.

Lo-fi cassette memory

1. Set **AGE** to 80 and **HISS** to 70.
2. Turn on **TAPE ECHO** with **WOW** at 90, **TONE** at 20 and **ECHO** at 25.
3. Turn on **CLOUD** in **REVERSE** mode with **MIX** at 35.

Reference

MIDI

Message	Effect
Note on/off	Plays and releases voices. Velocity is ignored, as on the original instruments.
Pitch bend	±2 semitones.
Sustain pedal (CC 64)	Holds released notes until the pedal comes up.
All Notes Off / All Sound Off	Releases every voice.

When all 8 voices are in use, a new note takes over the voice that started longest ago. Replaying a note that is still sounding retriggers the same voice.

Specifications

Item	Value
Polyphony	8 voices
Formats	VST3, Audio Unit, standalone app (macOS 11+, Apple Silicon and Intel)
Sample rates	Any the host runs at, such as 44.1, 48 or 96 kHz
Output	Stereo
Parameters	48, all automatable
Presets	12 factory, unlimited user
CPU	Around 2% of one core with all effects on at 48 kHz (measured on a desktop CPU)

Troubleshooting

- **Nimbus doesn't appear in Ableton.** Check that VST3 and/or Audio Units are turned on in **Settings → Plug-Ins**, then click **Rescan**. Look under **Plug-Ins → Javi**.
- **macOS blocks the installer.** Use **System Settings → Privacy & Security → Open Anyway** (see Installing).
- **There's a quiet hiss or hum when nothing is playing.** That's the HISS and AGE character, on purpose. Set **HISS** to 0 and **AGE** to 0 for silence.
- **The echoes keep growing. INTENS** is above about 85. Lower it, or switch TAPE ECHO off to clear the tape.
- **CLOUD keeps playing an old chord. FREEZE** is on. Turn it off.
- **Sound jumps when I change a preset.** Presets reset every control, including the effects. Save your own version first if you've made changes you want to keep.

The microKORG2 version

The Nimbus DCO also runs on the KORG microKORG2 (firmware 2.0 or later) as a logue SDK oscillator, `nimbus.mk2unit`, loaded with KORG's Librarian app. It has the 13 DCO, chorus and character controls; the microKORG2's own filter, envelopes and effects do the rest. Because the hardware gives each voice its own output, its chorus is mono per voice rather than the plug-in's stereo chorus.

The source code for both versions is at github.com/haveeay/nimbus-plugin.